

# BASSAM ARNAOUT

Sherbrooke, Canada

+1-438-725-8070

[bassamarnaout@outlook.com](mailto:bassamarnaout@outlook.com)

[linkedin.com/in/bassam-arnaout](https://linkedin.com/in/bassam-arnaout)

[github.com/bassamarnaout](https://github.com/bassamarnaout)

[youtube.com/@bassam-arnaout](https://youtube.com/@bassam-arnaout)

## CAREER SUMMARY

---

### Data Scientist • Machine Learning Engineer • AI Developer

Results-driven Data Scientist with a Master's degree in Computer Science (Machine Learning, AI, Data Mining) and 10+ years of software development experience. Skilled in building predictive models, deep learning pipelines, and scalable data-driven solutions. Strong statistical foundation with expertise in Python, SQL, ML algorithms, and end-to-end development. Proven ability to deliver measurable business impact through analytical problem-solving.

- Enhanced ML model accuracy by 12% through advanced feature engineering and hyperparameter tuning.
- Automated data preprocessing workflows, reducing manual processing time by 40%.

## CORE SKILLS

---

### Machine Learning & Data Science

- Supervised & Unsupervised Learning
- Feature Engineering & Model Optimization
- Classification, Regression, Clustering
- Image Processing & Computer Vision
- Statistical Modeling & Predictive Analytics

### Technical Skills

- Languages: Python, C#, C++, Java, SQL
- ML Tools: NumPy, Pandas, Scikit-learn, SciPy, XGBoost, MLP, SVM
- Deep Learning: Neural Networks, Hyperparameter Tuning
- Data Visualization: Tableau, Matplotlib
- Databases: MySQL, PostgreSQL, MS-SQL, MongoDB
- Cloud & Tools: Git, GitHub, REST APIs

### Software Engineering

- Full SDLC, Architecture, API Development
- Legacy System Modernization
- High-performance backend systems

## RELEVANT DATA SCIENCE & ML PROJECTS

---

### **Advanced Liver Fibrosis Prediction (ML & Deep Learning)** (GitHub: bassamarnaout)

Developed machine learning and deep learning models to predict advanced liver fibrosis in chronic Hepatitis C patients.

- Algorithms: LR, LDA, KNN, NB, CART, SVM, XGBoost, MLP
- Tasks: Preprocessing, feature extraction, model training, hyperparameter tuning
- Tools: Python, Scikit-learn, TensorFlow
- Achieved strong classification accuracy across multiple models

### **Predictive Loan Default Model** (GitHub: bassamarnaout)

Built a predictive model to estimate loan default probability using multiple classification algorithms.

- Used Logistic Regression, KNN, Gaussian NB, Decision Trees, SVM, Gradient Boosting, Extra Trees
- Conducted EDA, model comparison, and performance evaluation
- Improved prediction quality through feature selection

### **Image Analysis – Corrosion Detection:** (GitHub: bassamarnaout)

Implemented GLCM-based texture extraction and WCCD algorithm to identify corroded regions in industrial images.

- Techniques: Texture Analysis, Segmentation
- Technologies: Python, OpenCV, NumPy

### **Image Segmentation Using Mean Shift:** (GitHub: bassamarnaout)

Developed image segmentation algorithm using the mean-shift technique to separate color regions.

### **Social Media Usage Data Analysis (Tableau):**

Performed data cleaning, EDA, and dashboard creation to visualize behavior across social platforms.

## PROFESSIONAL EXPERIENCE

---

### **Senior Software Developer - Genatec Inc.** - Montreal, Quebec

**Nov 2020 – Present**

- Built Python and C#/.NET applications integrated with Sage 300 ERP, automating financial processes.
- Modernized legacy software, improving performance and maintainability.
- Delivered scalable backend solutions and contributed to architecture decisions.
- Improved customer satisfaction through effective debugging and support.

### **Customer Service Representative - Nordia (Canada Post)** - Sherbrooke, Quebec

**August 2020 – Nov 2020**

- Provided remote customer support with high communication standards.
- Awarded “Employee of the Week” 4 times for service excellence.

**Freelancer – Web Developer - Econoplus Electromenagers** - Sherbrooke, Quebec

**May 2020 – July 2020**

- Built an e-commerce platform and optimized UX for performance and navigation (Econoplus Électroménagers website).

**IT Project Manager / Software Developer - Mob2link** - Beirut, Lebanon

**Jan 2016 – August 2018**

- Led development of a Call Centre solution and the "Hayda Deal" mobile application.
- Managed product lifecycle, digital marketing, and technical delivery.
- Supervised the company Value Added Services portfolio based on market requirements & technology capabilities.
- Launched Intelligent Campaign Management System generating 9M SMS/month – Reduce Churn & Increase Usage-ARPU by 2.3%.

**IT Manager / Software Developer - Asaig** – Kuwait

**Jan 2014 – Dec 2015**

- Developed high-throughput SMS backend handling 555 SMS/sec using C#/.NET. Improved processing speed by 30%
- Led IT operations, infrastructure, and technical teams

## EDUCATION

**Master of Science (M.Sc.) in Computer Science - Bishop's University** - Sherbrooke, Canada.

**2018 - 2020**

Specialization: Machine Learning, Data Mining, Artificial Intelligence, Deep Learning, Neural Network.

**Bachelor (BE) in Electrical & Electronic Engineering - AUB (American University of Beirut)** – Lebanon

Focus: Computer & Communication Engineering Department CCE)

## CERTIFICATIONS & TRAINING

- Microsoft Certified Professional (MCP)
- MCSD.NET Training (New Horizon)
- PMP 35-hour training (Lifelong Learning)

## LANGUAGES

- English: Fluent
- French: Intermediate